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Optimal Energy-Saving Multi-Path Routing for Space-Air-Ground Integrated Networks

Lijun He, *Member, IEEE*, Yutong Li, Ziye Jia, *Member, IEEE*, Juncheng Wang, *Member, IEEE*, Minghui Min, *Member, IEEE*, and Zhu Han, *Fellow, IEEE*

***Abstract*—Traditional multi-path routing schemes for Space-Air-Ground Integrated Networks (SAGINs) pose the prominent challenge in reducing energy consumption. To address this, we study the energy minimization problem in multi-path routing by jointly optimizing link assignment, storage allocation, power control, and network routing in SAGINs over dynamic topologies. We first propose a Two-scale Time-Expanded Graph (TTEG) model to capture both network topology changes in large-timescale slots and mission data transmission in small-timescale slots. Using this TTEG model, we formulate the studied problem into a Mixed-Integer Non-Linear Program (MINLP) to minimize total energy consumption. We then leverage the inherent structure of the MINLP to divide it into two smaller subproblems. These two subproblems are solved iteratively with information feedback between them, thereby determining an optimal solution to the original problem in a few iteration steps. Simulation results validate the correctness and efficiency of the proposed solution.**

I. Introduction

Nowadays, Space-Air-Ground Integrated Networks (SAGINs) have been recognized as a critical component in establishing a global mobile communication network, with the potential to significantly enhance user experience in both densely populated areas and rural zones [1]. SAGINs integrate heterogeneous network nodes, including satellites, Unmanned Aerial Vehicles (UAVs), and terrestrial nodes, to provide global end-to-end transmission services via Inter-Satellite Links (ISLs) and Satellite-Ground Links (SGLs) [2]. The key to improving the quality of service in SAGINs lies in the design of an efficient routing strategy that effectively manages these heterogeneous network nodes and links.

Existing research on routing design for SAGINs falls into two categories: single-path routing and multi-path routing schemes. Single-path routing schemes [3], [4] only designate one path per mission from the source node to the destination node. This approach often struggles to guarantee reliable data transmission, particularly when a network node along the selected path fails. Furthermore, the single-path allocation constraint results in underutilization of the capacity of ISLs and SGLs in SAGINs. On the contrary, multi-path routing schemes make use of multiple paths per mission. These schemes fully exploit the presence of ISLs and SGLs in SAGINs by using time aggregated graph [5], ant colony optimization [6], and deep reinforcement learning [7]. Tang *et al.* [8] integrated traditional network coding with multi-path routing to facilitate coordination among multiple paths to enhance the network throughput of LEO satellite networks. Deng *et al.* [9] introduced a distance-based back-pressure routing scheme specifically designed to address the bursty nature of network traffic and the imbalanced regional communication loads in LEO satellite networks. However, existing multi-path routing schemes [5]–[9] fail to consider the impact of energy consumption on the performance of SAGINs. The unregulated utilization of onboard energy in satellite nodes may increase the frequency of recharging and discharging battery cells. This not only reduces the lifespan of the satellites, but may also result in routing interruptions. In this regard, Wang *et al.* [10] employed a time-expanded graph model to discretize the dynamic network topology into an adjacency matrix graph to minimize the total energy consumption of satellite and terrestrial integrated networks. However, this approach hinders the effectiveness of multi-path transmission strategy, particularly as the number of time slots and satellites increases. Thus, a major challenge in creating energy-efficient SAGINs lies in determining how to allocate multiple paths for each mission to minimize total energy consumption.

To this end, we study the energy minimization problem in multi-path routing through the joint optimization of link assignment, storage allocation, power control, and network routing in SAGINs over dynamic topologies. Our primary contributions include: We propose a novel Two-scale Time-Expanded Graph (TTEG) model to depict the dynamic properties of SAGINs, including both time-varying network topology in large-timescale slots and mission data transmission in small-timescale slots. Employing this TTEG model, we formulate the multi-path routing problem into a Mixed-Integer Non-Linear Program (MINLP) to minimize total energy consumption. Delving into the structure of the MINLP, we equivalently transform it into two smaller subproblems and then apply a feedback iteration method on them to obtain an optimal so-

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Lijun He and Minghui Min are with the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou 221116, China (e-mail: lijunhe@cumt.edu.cn; minmh@cumt.edu.cn). Lijun He is also with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an, Shaanxi, 710071, China.

Yutong Li is with the School of Software, Northwestern Polytechnical University, China (e-mail: yutongli.nwpu@gmail.com).

Ziye Jia is with the College of Electronic and Information Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, 211106, China (e-mail: jiaziye@nuaa.edu.cn).

Juncheng Wang is with the Department of Computer Science, Hong Kong Baptist University, China, (e-mail: jcwang@hkbu.edu.hk).

Zhu Han is with the Department of Electrical and Computer Engineering at the University of Houston, Houston, TX 77004 USA, and also with the Department of Computer Science and Engineering, Kyung Hee University, Seoul, 446-701, South Korea (e-mail: zhan2@uh.edu).

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lution to the original problem. Simulation results demonstrate that our proposed algorithm can find an optimal solution to the studied problem within a few iterations.

II. System Model and Problem Formulation

A. System Model

The primary factors influencing the dynamic topology of SAGIN are the rapid movements of LEO satellites, given that aerial nodes move at significantly slower speeds. To better concentrate on the dynamic properties of the SAGIN, our constructed network model excludes aerial nodes. Consider an SAGIN that comprises a set $\mathcal{I} = \{1, 2, ..., I\}$ of network nodes. The network nodes within the SAGIN consist of a set $\mathcal{R} = \{1, 2, ..., R\}$ of Data Relay Satellites (DRSs), a set $\mathcal{G} = \{1, 2, ..., G\}$ of Ground Stations (GSs), and a set $\mathcal{L} = \{1, 2, ..., L\}$ of Low Earth Orbit (LEO) satellites, i.e., $\mathcal{I} = \mathcal{R} \cup \mathcal{G} \cup \mathcal{L}$. Each node is equipped with a transmitter and receiver antenna, boasting a dual-function capability that allows it to operate both as a data collector and a data sink.

1) TTEG Model: In SAGIN, the mission data can be delivered between any two nodes only when they are within the coverage range of each other. We use time windows to model the coverage range of any node, which is a 2-tuple consisting of the start time and end time. The whole time axis is divided into non-overlapping intervals by these time windows [11]. As shown in Fig. 1, we define a large-timescale slot k as the time interval $[T_{k-1}, T_k)$, and all large-timescale slots constitute the set $\mathcal{K}$. Furthermore, we uniformly divide each large-timescale slot $k \in \mathcal{K}$ into a set of small-timescale slots with the same size, denoted by $\mathcal{T}_k$.

Based on the dual-timescale slots, we construct a TTEG $G_{\mathcal{T}}(\mathcal{V}, \mathcal{E}, \mathcal{W})$ to depict the network resources of SAGINs, where $\mathcal{V}$ denotes the set of vertexes, $\mathcal{E}$ represents the set of edges, and $\mathcal{W}$ indicates the set of weights. As illustrated in Fig. 1, the TTEG $G_{\mathcal{T}}(\mathcal{V}, \mathcal{E}, \mathcal{W})$ is a weighted directed graph spread across T layers, with each layer corresponding to the network topology within a large-timescale slot. Specifically, define $\mathcal{V} = \{v_i^k, i \in \mathcal{I}, k \in \mathcal{K}\}$, where vertex v_i^k signifies the network node i in large-timescale slot k. The edge set $\mathcal{E}$ comprises the communication edges $\mathcal{E}_{\text{Com}}$ and storage edges $\mathcal{E}_{\text{Stor}}$, i.e., $\mathcal{E} = \mathcal{E}_{\text{Com}} \cup \mathcal{E}_{\text{Stor}}$. We define $\mathcal{E}_{\text{Com}} = \{(v_i^k, v_j^k) | i \neq j \in \mathcal{I}, k \in \mathcal{K}\}$, where (v_i^k, v_j^k) denotes the available communication link between vertices v_i^k and v_j^k. Depending on the communication link types, we further categorize $\mathcal{E}_{\text{Com}}$ into $\mathcal{E}_{\text{Com}}^{\text{ISL}}$ and $\mathcal{E}_{\text{Com}}^{\text{SGL}}$, i.e., $\mathcal{E}_{\text{Com}} = \mathcal{E}_{\text{Com}}^{\text{ISL}} \cup \mathcal{E}_{\text{Com}}^{\text{SGL}}$. Here, $\mathcal{E}_{\text{Com}}^{\text{ISL}} = \{(v_i^k, v_j^k) | i \neq j \in \mathcal{R} \cup \mathcal{L}, k \in \mathcal{K}\}$ and $\mathcal{E}_{\text{Com}}^{\text{SGL}} = \{(v_i^k, v_j^k) | i \in \mathcal{R} \cup \mathcal{L}, j \in \mathcal{G}, k \in \mathcal{K}\}$ represent the edge set of ISLs and the edge set of SGLs, respectively. In addition, we define $\mathcal{E}_{\text{Stor}} = \{(v_i^k, v_i^{k+1}) | i \in \mathcal{I}, k \in \mathcal{K}\}$, where (v_i^k, v_i^{k+1}) represents the storage edge of vertex v_i^k. Furthermore, we define $\mathcal{C} = \{C_{v_i^k, v_j^k}(t)\}$ and $\mathcal{B} = \{B_{v_i^k}^{\max}\}$, such that $\mathcal{W} = \mathcal{C} \cup \mathcal{B}$, where $\mathcal{C}$ and $\mathcal{B}$ are the sets of weights on communication and storage links, respectively. Specifically, $C_{v_i^k, v_j^k}(t)$ represents the channel capacity of the communication edge (v_i^k, v_j^k) in a small-timescale slot t, while $B_{v_i^k}^{\max}$ represents the maximum storage space of vertex v_i^k.

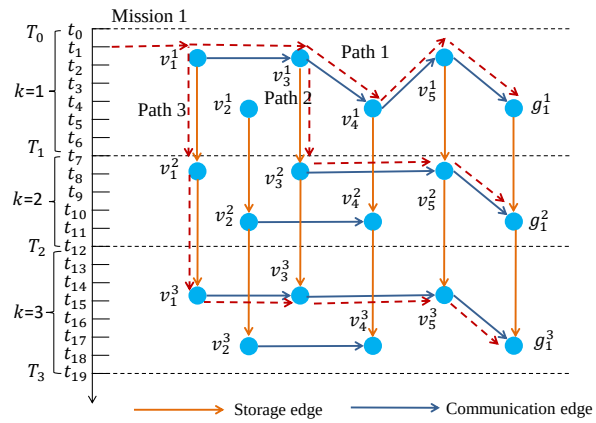

Fig. 1. An illustration of the proposed TTEG model.

Remark 1. Different from the conventional time-varying graph models [12] that employ a fixed time slot division approach, the proposed TTEG introduces a novel dual-scale time slot division strategy. Specifically, the large-timescale slots are utilized to capture the temporal evolution of the SAGIN network topology, while aggregating the topological changes across multiple small-timescale slots. This dual-scale time slot structure markedly reduces the number of nodes and edges in the TTEG, thereby greatly simplifying the complexity of problem modeling and enhancing modeling efficiency.

2) Communication Model: The channel capacity of any ISL $(v_i^k, v_j^k) \in \mathcal{E}_{\text{Com}}^{\text{ISL}}$ is expressed as

$$C_{v_i^k, v_j^k}(t) = g_{v_i^k, v_j^k}^{\text{ISL}}(t) P_{v_i^k, v_j^k}(t), \forall t \in \mathcal{T}_k, k, \tag{1}$$

where $P_{v_i^k, v_j^k}(t)$ is the transmit power of vertex v_i^k in small-timescale slot t and $g_{v_i^k, v_j^k}^{\text{ISL}}(t)$ is the channel gain of the ISL (v_i^k, v_j^k) in small-timescale slot t, given by

$$g_{v_i^k, v_j^k}^{\text{ISL}}(t) = \frac{G_{v_i^k}^{\text{tran}}(t) G_{v_j^k}^{\text{rec}}(t) L_f(t) L_l(t)}{\kappa T_s (E_b/N_0)_{\text{req}} M}, \forall t \in \mathcal{T}_k, k. \tag{2}$$

In (2), $G_{v_i^k}^{\text{tran}}(t)$ denotes the transmit antenna gain of vertex v_i^k in small-timescale slot t, $G_{v_j^k}^{\text{rec}}(t)$ represents the data receiver antenna gain of vertex v_j^k in small-timescale slot t, $L_f(t)$ is the free space loss in small-timescale slot t, $L_l(t)$ is the total line loss in small-timescale slot t, κ is the Boltzmann's constant (in JK^{-1}), T_s is the total system noise temperature (in K), $(E_b/N_0)_{\text{req}}$ is the required ratio of received energy-per-bit to noise-density, and M is the link margin. The channel capacity of any SGL $(v_i^k, v_j^k) \in \mathcal{E}_{\text{Com}}^{\text{SGL}}$ is computed as follows:

$$C_{v_i^k, v_j^k}(t) = B_c \log_2 \left(1 + \gamma_{v_i^k, v_j^k}(t)\right), \forall t \in \mathcal{T}_k, k, \tag{3}$$

where B_c is the link bandwidth and $\gamma_{v_i^k, v_j^k}(t)$ is the signal-to-noise ratio of SGL (v_i^k, v_j^k) in small-timescale t, given by

$$\gamma_{v_i^k, v_j^k}(t) = g_{v_i^k, v_j^k}^{\text{SGL}}(t) P_{v_i^k}(t), \quad \forall t \in \mathcal{T}_k, k. \tag{4}$$

In (4), $g_{v_i^k, v_j^k}^{\text{SGL}}(t)$ is the channel gain of the SGL (v_i^k, v_j^k) in small-timescale slot t, calculated by

$$g_{v_i^k, v_j^k}^{\text{SGL}}(t) = \frac{G_{v_i^k}^{\text{tran}}(t) G_{v_j^k}^{\text{rec}}(t) L_f(t) L_l(t)}{N_0}, \forall t \in \mathcal{T}_k, k, \tag{5}$$

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where N_0 is the noise power.

3) Energy Consumption Model: The total energy consumption of SAGINs is calculated by

$$E^{\text{total}} = E^{\text{static}}_{\text{total}} + E^{\text{dyn}}_{\text{total}}, \tag{6}$$

where $E^{\text{static}}_{\text{total}}$ is the total static energy consumption and $E^{\text{dyn}}_{\text{total}}$ is the total dynamic energy consumption. We introduce variable $y_{v_i^k,v_j^k}(t) \in \{0,1\}$ to represent the link assignment strategy, such that $y_{v_i^k,v_j^k}(t) = 1$ indicates that link $(v_i^k, v_j^k) \in \mathcal{E}$ is selected in small-timescale slot $t \in \mathcal{T}_k$; otherwise $y_{v_i^k,v_j^k}(t) = 0$. The total static energy consumption is calculated by

$$E^{\text{static}}_{\text{total}} = \sum_{\substack{k\in\mathcal{K}\\ t\in\mathcal{T}_k}} \sum_{(v_i^k,v_j^k)\in\mathcal{E}} y_{v_i^k,v_j^k}(t) E^{\text{static}}_{v_i^k,v_j^k}(t), \tag{7}$$

where $E^{\text{static}}_{v_i^k,v_j^k}(t)$ is the static energy consumption between v_i^k and v_j^k in any small-timescale slot $t \in \mathcal{T}_k$. The static energy consumption $E^{\text{static}}_{v_i^k,v_j^k}(t)$ is the energy required to sustain communication and normal operations, which is typically independent of mission data transmission [13]. Consequently, the value of $E^{\text{static}}_{v_i^k,v_j^k}(t)$ is considered as a constant with respect to v_i^k in any small-timescale slot t. The total dynamic energy consumption can be evaluated by

$$E^{\text{dyn}}_{\text{total}} = \sum_{\substack{k\in\mathcal{K}\\ t\in\mathcal{T}_k}} \sum_{(v_i^k,v_j^k)\in\mathcal{E}} y_{v_i^k,v_j^k}(t) E^{\text{dyn}}_{v_i^k,v_j^k}(t), \tag{8}$$

where $E^{\text{dyn}}_{v_i^k,v_j^k}(t)$ is the dynamic energy consumption of the mission data transmission from v_i^k to v_j^k in any small-timescale slot $t \in \mathcal{T}_k$, expressed by

$$E^{\text{dyn}}_{v_i^k,v_j^k}(t) = P_{v_i^k,v_j^k}(t)\tau, \quad \forall (v_i^k, v_j^k), t \in \mathcal{T}_k. \tag{9}$$

4) Task Model: Denote a set of missions as $\mathcal{M} = \{1, 2, ..., M\}$. Each mission $m \in \mathcal{M}$ is defined as a 5-tuple $(u_m, D_m, t_m, d_m, g_m)$, where u_m is the data collector to produce mission m, D_m is the data volume of mission m, t_m is the arrival time of mission m, d_m is the deadline of mission m, and g_m is the destination node of mission m. Any mission m is successfully completed if it can be modeled as a flow f_m which is sent out by data collector u_m at the arrival time t_m and received by data sink g_m before its deadline d_m through multi-path route. Define $\lambda^{f_m}_{v_i^k,v_j^k}(t) \geq 0$ as the portion of the data volume D_m of flow f_m on edge (v_i^k, v_j^k) in small-timescale slot t. Let $\mathcal{F} = \{f_m\}$ denote the set of all flows.

B. Problem Formulation

The following constraint ensures that a maximum of one data receiving vertex is selected to receive the data from any data sending vertex in any small-timescale slot:

$$\sum_{v_j^k:(v_i^k,v_j^k)\in\mathcal{E}_{\text{Com}}} y_{v_i^k,v_j^k}(t) \leq 1, \quad \forall v_i^k, t \in \mathcal{T}_k, k. \tag{10}$$

Similarly, constraint (11) guarantees that at most one data sending vertex is selected to transfer its data to any data receiving vertex within any small-timescale slot:

$$\sum_{v_i^k:(v_i^k,v_j^k)\in\mathcal{E}_{\text{Com}}} y_{v_i^k,v_j^k}(t) \leq 1, \quad \forall v_j^k, t \in \mathcal{T}_k, k. \tag{11}$$

Constraint (12) indicates that a vertex cannot simultaneously transmit and receive data, i.e., for any $v_i^k, t \in \mathcal{T}_k, k$,

$$\sum_{v_j^k:(v_i^k,v_j^k)\in\mathcal{E}_{\text{Com}}} y_{v_i^k,v_j^k}(t) + \sum_{v_j^k:(v_j^k,v_i^k)\in\mathcal{E}_{\text{Com}}} y_{v_j^k,v_i^k}(t) \leq 1. \tag{12}$$

Furthermore, constraint (13) specifies the limitation of communication capacity, i.e., for any edge $(v_i^k, v_j^k), t \in \mathcal{T}_k, k$,

$$\sum_{f_m\in\mathcal{F}} \lambda^{f_m}_{v_i^k,v_j^k}(t) D_m \leq y_{v_i^k,v_j^k}(t) C_{v_i^k,v_j^k}(t)\tau. \tag{13}$$

Constraint (14) enforces that the total volume of data for any flow f_m should be produced by the collector u_m in slot t_m:

$$b^t_{v_i^k}(f_m) = 1, \forall f_m, v_i^k = u_m, t = t_m, \tag{14}$$

where $b^t_{v_i^k}(f_m) \geq 0$ is a variable to represent the fraction of storage data amount of vertex v_i^k in slot t. Similarly, constraint (15) reflects that all the data amount of any flow f_m should be delivered to its destination g_m before deadline d_m, i.e.,

$$b^t_{v_i^k}(f_m) = 1, \forall f_m, v_i^k = g_m, t = d_m. \tag{15}$$

The flow conservation constraint is given by

$$\begin{aligned}&\sum_{v_j^k:(v_j^k,v_i^k)\in\mathcal{E}} \lambda^{f_m}_{v_j^k,v_i^k}(t) - \sum_{v_j^k:(v_i^k,v_j^k)\in\mathcal{E}} \lambda^{f_m}_{v_i^k,v_j^k}(t)\\ &= b^t_{v_i^k}(f_m) - b^{t-1}_{v_i^k}(f_m), \forall v_i^k, f_m, t \in \Gamma_k(f_m),\end{aligned} \tag{16}$$

where $\Gamma_k(f_m) = [t_m + 1, d_m] \bigcap \mathcal{T}_k$. Constraint (17) demonstrates the storage capacity limitation of any given vertex:

$$0 \leq \sum_{f_m\in\mathcal{F}} b^t_{v_i^k}(f_m) D_m \leq B^{\max}_{v_i^k}, \quad \forall v_i^k, t \in \mathcal{T}_k, k. \tag{17}$$

Moreover, constraint (18) reveals that the storage and release amount of v_i^k should be respectively less than its inflow and outflow amount, i.e.,

$$\begin{aligned}&-\sum_{v_j^k:(v_i^k,v_j^k)\in\mathcal{E}_{\text{Com}}} y_{v_i^k,v_j^k}(t) \leq b^t_{v_i^k}(f_m) - b^{t-1}_{v_i^k}(f_m)\\ &\leq \sum_{v_j^k:(v_j^k,v_i^k)\in\mathcal{E}_{\text{Com}}} y_{v_j^k,v_i^k}(t), \quad \forall v_i^k, f_m, t \in \mathcal{T}_k, k.\end{aligned} \tag{18}$$

The following constraint reflects the range of transmit power control :

$$P_{\min} \leq P_{v_i^k,v_j^k}(t) \leq P_{\max}, \forall (v_i^k, v_j^k) \in \mathcal{E}_{\text{Com}}, t \in \mathcal{T}_k, k, \tag{19}$$

where $P_{\min}$ and $P_{\max}$ denote the minimum transmit power and the maximum transmit power, respectively. Constraint (20) ensures that the data amount of any mission is unchanged:

$$\sum_{v_i^k\in\mathcal{V}} b^t_{v_i^k}(f_m) = 1, \quad \forall f_m, k, t \in \digamma_k(f_m), \tag{20}$$

where $\digamma_k(f_m) = [t_m + 1, \mathcal{T}_k]$

Based on the aforementioned constraints, we formulate the studied problem as the following form:

$$\begin{aligned}\textbf{P0}:\ &\min_{\boldsymbol{y},\boldsymbol{b},\boldsymbol{P},\boldsymbol{\lambda}} \quad E^{\text{total}}\\ &\text{s.t. (10)-(20)},\\ &y_{v_i^k,v_j^k}(t) \in \{0,1\}, \forall (v_i^k, v_j^k) \in \mathcal{E}_{\text{Com}}, t \in \mathcal{T}_k, k,\end{aligned} \tag{21}$$

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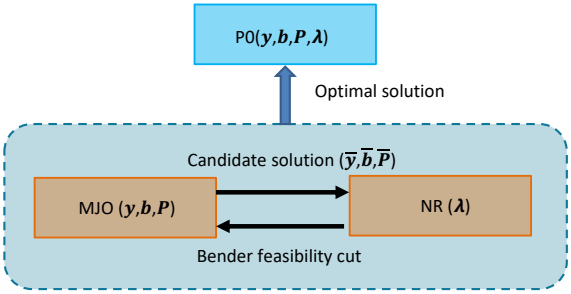

Fig. 2. An illustration of the information feedback mechanism between MJO and NR problems.

$$0 \le \lambda^{f_m}_{v_i^k,v_j^k}(t) \le 1, \forall v_i^k \ne v_j^k, f_m, t \in \mathcal{T}_k, k, \tag{22}$$

$$0 \le b^t_{v_i^k}(f_m) \le 1, \forall v_i^k, f_m, t \in \mathcal{T}_k, k, \tag{23}$$

where $\boldsymbol{y} = \{y_{v_i^k,v_j^k}(t)\}$, $\boldsymbol{b} = \{b^t_{v_i^k}(f_m)\}$, $\boldsymbol{P} = \{P_{v_i^k,v_j^k}(t)\}$, and $\boldsymbol{\lambda} = \{\lambda^{f_m}_{v_i^k,v_j^k}(t)\}$.

Remark 2. Since it is a MINLP, **P0** is challenging to solve optimally in a low computation cost. In particular, the variables of $\boldsymbol{y}$, $\boldsymbol{\lambda}$, and $\boldsymbol{P}$ are coupled in both the objective function and the nonlinear constraint (12), further complicating the problem solving. In addition, the objective function of **P0** incorporates the term of $E^{\text{static}}_{\text{total}}$, which composed of a binary variable $\boldsymbol{y}$ and a constant with respect to v_i^k in any small-timescale slot t. This further expands the combinatorial optimization space of **P0**. To this end, we explore the unique structure of the formulation to reduce the size of solution space aiming to devise an efficient algorithm for the large-scale problem **P0**.

III. Optimal Energy-saving Multi-path Routing

From **P0**, we observe two insights for facilitating problem decomposition. The first is that $\boldsymbol{\lambda}$ is not in the objection function of **P0**. The second is that $\boldsymbol{\lambda}$ is constrained by $(\boldsymbol{y}, \boldsymbol{P})$ in (13) and $\boldsymbol{b}$ in (16). These two insights motivate us to divide $(\boldsymbol{y}, \boldsymbol{b}, \boldsymbol{P}, \boldsymbol{\lambda})$ into two categories of $(\boldsymbol{y}, \boldsymbol{b}, \boldsymbol{P})$ and $\boldsymbol{\lambda}$. Correspondingly, **P0** can be equivalently decomposed into the Multi-resource Joint Optimization (MJO) problem and the Network Routing (NR) problem. These two relatively "easy" problems are iteratively solved through the information feedback between them until find an optimal solution to **P0**, as shown in Fig. 2. In detial, the MJO problem is to simultaneously optimize the communication links, storage resources, and power resources to produce a candidate solution $(\bar{\boldsymbol{y}}, \bar{\boldsymbol{b}}, \bar{\boldsymbol{P}})$ for the NR problem. Upon obtaining this candidate solution $(\bar{\boldsymbol{y}}, \bar{\boldsymbol{b}}, \bar{\boldsymbol{P}})$, the responsibility of the NR problem is to find a feasible multi-path routing for missions. It can either (i) confirm that $(\bar{\boldsymbol{y}}, \bar{\boldsymbol{b}}, \bar{\boldsymbol{P}})$ represents an optimal solution for **P0** or (ii) provide a Bender feasibility cut for the MJO problem, thereby excluding $(\bar{\boldsymbol{y}}, \bar{\boldsymbol{b}}, \bar{\boldsymbol{P}})$ from the feasible set.

For any fixed $\bar{\boldsymbol{\lambda}}$, the residual problem of **P0** is written as

$$\text{MJO:} \quad \min_{\boldsymbol{y},\boldsymbol{b},\boldsymbol{P}} \quad E^{\text{total}}$$
$$\text{s.t. } (10)\text{-}(12), (14)\text{-}(15), (17)\text{-}(21), (23).$$

The dimension of the MJO problem is significantly smaller than that of **P0**, since it has less variables and constraints. This motivates us to use commercial solvers (e.g., CVX) to solve it achieving an optimal solution $(\bar{\boldsymbol{y}}, \bar{\boldsymbol{b}}, \bar{\boldsymbol{P}})$. However, $(\bar{\boldsymbol{y}}, \bar{\boldsymbol{b}}, \bar{\boldsymbol{P}}, \bar{\boldsymbol{\lambda}})$ is not necessarily the optimal solution to **P0**, unless there exists a feasible solution $\bar{\boldsymbol{\lambda}}$ that satisfies the constraints of (13), (16), and (22). Thus, we require solving the NR problem to verify whether $(\bar{\boldsymbol{y}}, \bar{\boldsymbol{b}}, \bar{\boldsymbol{P}}, \bar{\boldsymbol{\lambda}})$ is an optimal solution of **P0**.

For a given $(\bar{\boldsymbol{y}}, \bar{\boldsymbol{b}}, \bar{\boldsymbol{P}})$, **P0** degrades into the following form:

$$\text{NR:} \quad \min_{\boldsymbol{\lambda}} \quad 0$$
$$\text{s.t. } (22),$$
$$\sum_{f_m \in \mathcal{F}} \lambda^{f_m}_{v_i^k,v_j^k}(t) D_m \le \bar{y}_{v_i^k,v_j^k}(t) C_{v_i^k,v_j^k}(t)\tau,$$
$$\forall (v_i^k, v_j^k) \in \mathcal{E}, t \in \mathcal{T}_k, k, \tag{24}$$
$$\sum_{v_j^k:(v_j^k,v_i^k)\in\mathcal{E}} \lambda^{f_m}_{v_j^k,v_i^k}(t) - \overline{b^{t-1}_{v_i^k}}(f_m) = \overline{b^t_{v_i^k}}(f_m)$$
$$- \sum_{v_j^k:(v_i^k,v_j^k)\in\mathcal{E}} \lambda^{f_m}_{v_i^k,v_j^k}(t), \quad \forall v_i^k, f_m, t \in \Gamma_k(f_m). \tag{25}$$

The NR problem is easy to solve, since it is a standard Linear Program (LP). If finding a feasible solution $\bar{\boldsymbol{\lambda}}$ satisfying the constraints of (22), (24), and (25) with the given $(\bar{\boldsymbol{y}}, \bar{\boldsymbol{b}}, \bar{\boldsymbol{P}})$, then $(\bar{\boldsymbol{y}}, \bar{\boldsymbol{b}}, \bar{\boldsymbol{P}}, \bar{\boldsymbol{\lambda}})$ is an optimal solution to **P0**; otherwise, we utilize the NR problem with the known $(\bar{\boldsymbol{y}}, \bar{\boldsymbol{b}}, \bar{\boldsymbol{P}})$ to construct the feedback information of Bender feasibility cut. To this end, we first define the Lagrangian function of $L(\boldsymbol{\lambda}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\delta})$ associated with the NR problem with $\boldsymbol{\alpha} = \{\alpha_{v_i^k,v_j^k}(t)\}$, $\boldsymbol{\beta} = \{\beta^{f_m}_{v_i^k(t)}(t)\}$, and $\boldsymbol{\delta} = \{\delta^{f_m}_{v_i^k,v_j^k}(t)\}$ as follows:

$$L(\boldsymbol{\lambda}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\delta}) = \sum_{\substack{k\in\mathcal{K}\\ t\in\mathcal{T}_k}} \sum_{(v_i^k,v_j^k)\in\mathcal{E}} \alpha_{v_i^k,v_j^k}(t)\theta^t_{v_i^k,v_j^k}$$
$$+ \sum_{\substack{t\in\Gamma_k(f_m)\\ f_m\in\mathcal{F}}} \sum_{v_i^k\in\mathcal{V}} \beta^{f_m}_{v_i^k}(t)\mu^{f_m,t}_{v_i^k} + \sum_{\substack{k\in\mathcal{K}\\ t\in\mathcal{T}_k}} \sum_{\substack{(v_i^k,v_j^k)\in\mathcal{E}\\ f_m\in\mathcal{F}}} \psi^{f_m,t}_{v_i^k,v_j^k}, \tag{26}$$

where $\theta^t_{v_i^k,v_j^k} = \sum_{f_m\in\mathcal{F}} \lambda^{f_m}_{v_i^k,v_j^k}(t) D_m - \bar{y}_{v_i^k,v_j^k}(t) C_{v_i^k,v_j^k}(t)\tau$, $\mu^{f_m,t}_{v_i^k} = \sum_{v_j^k:(v_j^k,v_i^k)\in\mathcal{E}} \lambda^{f_m}_{v_j^k,v_i^k}(t) - \sum_{v_j^k:(v_i^k,v_j^k)\in\mathcal{E}} \lambda^{f_m}_{v_i^k,v_j^k}(t) - \overline{b^t_{v_i^k}}(f_m) + \overline{b^{t-1}_{v_i^k}}(f_m)$, and $\psi^{f_m,t}_{v_i^k,v_j^k} = \delta^{f_m}_{v_i^k,v_j^k}(t)(\lambda^{f_m}_{v_i^k,v_j^k}(t) - 1)$. Then, we give the following constraint:

$$\alpha_{v_i^k,v_j^k}(t) D_m + \gamma^{f_m}_{v_j^k,v_i^k}(t) + \delta^{f_m}_{v_i^k,v_j^k}(t) \ge 0,$$
$$\forall (v_i^k, v_j^k) \in \mathcal{E}, t \in \mathcal{T}_k, k, f_m, \tag{27}$$

where $\gamma^{f_m}_{v_j^k,v_i^k}(t) = \beta^{f_m}_{v_j^k}(t) - \beta^{f_m}_{v_i^k}(t)$ for $t \in \Gamma_k(f_m)$ and otherwise $\gamma^{f_m}_{v_j^k,v_i^k}(t) = 0$. Based on constraint (27), we give the Lagrangian dual function of $L(\boldsymbol{\lambda}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\delta})$ as follows:

$$g(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\delta}) = \min_{\boldsymbol{\lambda}\ge 0} L(\boldsymbol{\lambda}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\delta}) = - \sum_{\substack{k\in\mathcal{K}\\ t\in\mathcal{T}_k}} \sum_{\substack{(v_i^k,v_j^k)\in\mathcal{E}\\ f_m\in\mathcal{F}}} \delta^{f_m}_{v_i^k,v_j^k}(t)$$
$$- \sum_{\substack{t\in\Gamma_k(f_m)\\ f_m\in\mathcal{F}}} \sum_{v_i^k\in\mathcal{V}} \beta^{f_m}_{v_i^k}(t)(\overline{b^t_{v_i^k}}(f_m) - \overline{b^{t-1}_{v_i^k}}(f_m))$$
$$- \sum_{\substack{k\in\mathcal{K}\\ t\in\mathcal{T}_k}} \sum_{(v_i^k,v_j^k)\in\mathcal{E}} \alpha_{v_i^k,v_j^k}(t)\bar{y}_{v_i^k,v_j^k}(t) C_{v_i^k,v_j^k}(t)\tau,$$

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when $(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\delta}) \in \mathcal{A} = \{(27), \boldsymbol{\alpha} \geq 0, \boldsymbol{\delta} \geq 0\}$. If $(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\delta}) \notin \mathcal{A}$, then $g(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\delta}) = -\infty$. Furthermore, we obtain the dual problem of the NR problem as follows:

$$\text{NR-D:} \quad \max_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\delta}} \quad g(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\delta}) \quad \text{s.t.} \quad (\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\delta}) \in \mathcal{A}.$$

The NR problem has a feasible solution if and only if the optimal objective value to the NR-D problem is nonnegative, i.e., $g^*(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\delta}) \geq 0$. It is easy to verify that all-zero vector is a feasible solution to the NR-D problem. Thus, the optimal objective value $g^*(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\delta})$ to the NR-D problem is either zero or unbounded. If $g^*(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\delta}) = 0$, then there exists a feasible solution $\bar{\boldsymbol{\lambda}}$ for the NR problem and an optimal solution to **P0** is $(\bar{\boldsymbol{y}},\bar{\boldsymbol{b}},\bar{\boldsymbol{P}},\bar{\boldsymbol{\lambda}})$. If $g^*(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\delta}) = -\infty$, then there exists an extreme ray $(\overline{\boldsymbol{\alpha}}, \overline{\boldsymbol{\beta}}, \overline{\boldsymbol{\delta}})$ of the NR-D problem to satisfy:

$$\begin{aligned}\eta(\boldsymbol{y},\boldsymbol{b},\boldsymbol{P}) &= \sum_{\substack{k\in\mathcal{K}\\ t\in\mathcal{T}_k}} \sum_{(v_i^k,v_j^k)\in\mathcal{E}} \overline{\alpha}_{v_i^k,v_j^k}(t) y_{v_i^k,v_j^k}(t) C_{v_i^k,v_j^k}(t)\tau \\ &+ \sum_{\substack{t\in\Gamma_k(f_m)\\ f_m\in\mathcal{F}}} \sum_{v_i^k\in\mathcal{V}} \overline{\beta}_{v_i^k}^{f_m}(t)\left(b_{v_i^k}^t(f_m) - b_{v_i^k}^{t-1}(f_m)\right) \\ &+ \sum_{\substack{k\in\mathcal{K}\\ t\in\mathcal{T}_k}} \sum_{\substack{(v_i^k,v_j^k)\in\mathcal{E}\\ f_m\in\mathcal{F}}} \overline{\delta}_{v_i^k,v_j^k}^{f_m}(t) < 0. \end{aligned} \tag{28}$$

The inequality of (28) indicates that $(\bar{\boldsymbol{y}},\bar{\boldsymbol{b}},\bar{\boldsymbol{P}})$ is unable to define a feasible solution of **P0**. This motivate us to construct the following linear inequality (i.e., a Bender feasibility cut) to remove $(\bar{\boldsymbol{y}},\bar{\boldsymbol{b}},\bar{\boldsymbol{P}})$ from the feasible set of the MJO problem:

$$\eta(\boldsymbol{y},\boldsymbol{b},\boldsymbol{P}) \geq 0. \tag{29}$$

Obviously, inequality (29) always holds for **P0**. Thus, adding (29) into the MJO problem can further reduce its feasible set. The above information feedback procedure is repeated until finding an optimal solution to **P0**. We refer to the aforementioned solution as the Optimal Energy-saving Multi-path Routing (OEMR) and summarize it in Algorithm 1. The computational complexity of the proposed OEMR algorithm is primarily determined by three key steps: solving the MJO problem in step 4, the NR problem in step 7, and the NR-D problem in step 11. The MJO problem involves $\Psi = (\sum_{k\in\mathcal{K}} |\mathcal{T}_k|(2|\mathcal{E}_{\text{Com}}| + |\mathcal{F}||\mathcal{V}|)$ decision variables and $\Phi = \sum_{k\in\mathcal{K}} |\mathcal{T}_k|(|\mathcal{V}| + |\mathcal{E}| + 2(|\mathcal{E}_{\text{Com}}|+|\mathcal{F}|+|\mathcal{F}||\mathcal{V}|)) + \sum_{f_m\in\mathcal{F}} \sum_{k\in\mathcal{K}} (|\Gamma_k(f_m)||\mathcal{V}|+|\Gamma_k(f_m)|)$ constraints. Consequently, the computational complexity of solving the MJO problem to obtain ϵ-optimal solutions using Gurobi is $O(\sqrt{\Phi}\ln(\frac{1}{\epsilon})\Psi(\Phi + \Psi\Phi + \Psi^2)$. The subproblems of NR and NR-D are standard LP problems, and their computational complexities are $O((\sum_{k\in\mathcal{K}} |\mathcal{T}_k||\mathcal{E}|(|\mathcal{F}| + 1) + \sum_{f_m\in\mathcal{F}} \sum_{k\in\mathcal{K}} |\Gamma_k(f_m)||\mathcal{V}|)^{3.5})$ and $O((\sum_{k\in\mathcal{K}} |\mathcal{T}_k||\mathcal{F}||\mathcal{E}|)^{3.5})$. Therefore, the computational complexity of the OEMR algorithm is $O((\sqrt{\Phi}\ln(\frac{1}{\epsilon})\Psi(\Phi + \Psi\Phi + \Psi^2) + (\sum_{k\in\mathcal{K}} |\mathcal{T}_k||\mathcal{E}|(|\mathcal{F}| + 1) + \sum_{f_m\in\mathcal{F}} \sum_{k\in\mathcal{K}} |\Gamma_k(f_m)||\mathcal{V}|)^{3.5})N_{\max})$, where N^{Max} denotes the maximum number of iterations.

Algorithm 1 Proposed OEMR Algorithm

1: Set $\mathcal{C} \leftarrow \varnothing$, $N \leftarrow 1$.
2: **while** $N \leq N_{\max}$ **do**
3: Add $\mathcal{C}$ into the MJO problem.
4: Solve the MJO problem to obtain $(\bar{\boldsymbol{y}},\bar{\boldsymbol{b}},\bar{\boldsymbol{P}})$.
5: Use $(\bar{\boldsymbol{y}},\bar{\boldsymbol{b}},\bar{\boldsymbol{P}})$ to formulate the NR problem.
6: **if** the NR problem is feasible **then**
7: Solve the NR problem to obtain $\bar{\boldsymbol{\lambda}}$.
8: Output the optimal solution $(\bar{\boldsymbol{y}},\bar{\boldsymbol{b}},\bar{\boldsymbol{P}},\bar{\boldsymbol{\lambda}})$.
9: $N \leftarrow N_{\max} + 1$.
10: **else**
11: Solve the NR-D problem to obtain $(\bar{\boldsymbol{\alpha}},\bar{\boldsymbol{\beta}},\bar{\boldsymbol{\delta}})$.
12: Use $(\bar{\boldsymbol{\alpha}},\bar{\boldsymbol{\beta}},\bar{\boldsymbol{\delta}})$ to construct (29).
13: $\mathcal{C} \leftarrow= \mathcal{C} \cup \{(29)\}$.
14: $N \leftarrow N + 1$.
15: **end if**
16: **end while**

IV. Simulation Evaluation

The simulation scenario comprises three DRSs, three LEO satellites, and one GS. The scheduling horizon is set from 9 May 2024 21:00:00.0 to 9 May 2024 21:40:00.0. The three DRSs lie at nominal longitudes of 0°E, 120°E, and 120°W. The orbit altitudes, inclination angles, and local time of ascending nodes for the LEO satellites are 550km, 53°, and 21:00. The location of GS is set to (39.6°N, 115.9°E). The data amount for each mission is randomly generated in $[15, 30]$Mbit. The value of $B_{v_i^k}^{\max}, \forall v_i^k$ is randomly generated within $[150, 300]$Mbit. We set $P_{\min} = 100$W, $P_{\max} = 200$W, $G_{v_i^{T_k}}^{\text{tran}}(t) = 37$dB, $G_{v_i^{T_k}}^{\text{rec}}(t) = 37$dB, $M = 1.4125$dB, $L_f(t) = 10^{-23}$, $L_l(t) = 1$, $\kappa = 1.38 * 10^{23}\text{JK}^{-1}$, $T_s = 316.2278$K, $(E_b/N_0)_{\text{req}} = 9.6$dB, and $B_c = 4 * 10^{10}$Hz. To solve the MJO problem and the NR problem, we use Anaconda to call Gurobi 11.0's solver. We utilize the software of Matlab to call the CVX toolbox to solve **P0** with a fixed $\boldsymbol{P}$.

In Fig. 3a, we show how the total energy consumption varies with the increasing transmit power for different values ranges of the static energy consumption. From Fig. 3a, we observe that the total energy consumption fluctuates as the transmit power increasing, when transmit power is less than 110W. The total energy consumption is reduced, due to the fact that the capacity of ISLs and SGLs enlarges with the increasing transmit power to yield some feasible routing paths with the smaller static energy consumption. The total energy consumption is increased, since the dynamic energy consumption significantly increases as the transmit power grows. Furthermore, it is observed from Fig. 3a that the total energy consumption monotonically increases as the transmit power increases, when the transmit power is larger than 110W. The main reason is that the part of the total energy consumption increased by the dynamic energy consumption is larger than the part decreased by the routing path selection. Moreover, it can be seen from Fig. 3a that the large value of the static energy consumption achieves the larger total energy consumption as the transmit power growing. This is because the total energy consumption increases with the increasing static energy consumption when

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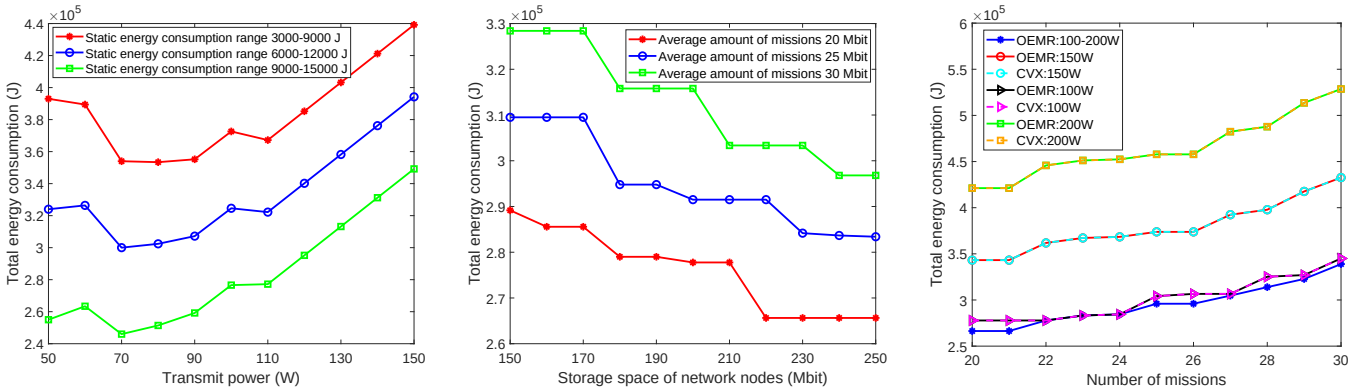

(a) The transmit power. (b) The storage space of network nodes. (c) The number of missions.

Fig. 3. Total energy consumption versus various network parameters.

the dynamic energy consumption remains unchanged.

Fig. 3b plots the total energy consumption versus the storage space of network nodes for the different average amount of missions. From Fig. 3b, we observe that the total energy consumption first decreases and converges to a minimization value with the increased storage space of the network nodes. This is because the network nodes with the larger storage space can store more mission data until finding some feasible routing paths with the smaller total energy consumption. The total energy consumption eventually does not change, since there is not feasible lower-energy consumption routing paths. Furthermore, we observe from Fig. 3b that the larger average mission amount consumes the larger total energy consumption than the smaller one. The reason is that the larger total energy consumption needs to route the more space data, when the average mission amount is large.

In Fig. 3c, we show the performance comparisons between the proposed OEMR algorithm and CVX in terms of the total energy consumption versus the number of missions for the different values of transmit power. For the fairness of comparison, we require fixing the value of $\boldsymbol{P}$ in **P0**, since the toolbox of CVX cannot solve it directly. From Fig. 3c, it can be seen that the total energy consumption of the proposed OEMR algorithm is the same as that of CVX for the different values of the fixed $\boldsymbol{P}$. This phenomenon illustrates the proposed OEMR algorithm can obtain an optimal solution to **P0** with fixing $\boldsymbol{P}$. Furthermore, the proposed OEMR algorithm with a transmit power range of $[100, 200]$W achieves a lowest total energy consumption than the alternatives. This observation indicates that joint optimization of power control and network routing facilitates the total energy consumption reduction of SAGINs.

V. Conclusion

In this paper, we have studied multi-path routing for SAGINs to minimize total energy consumption, subject to the constraints of link capacity and node storage. Specifically, we propose a new TTEG model to characterize the dynamic properties of SAGINs, including the dynamic network topology in large-timescale slots and mission data transmission in small-timescale slots. Using the TTEG model, we formulate the studied problem as an MINLP and exploit its special structure to propose an efficient OEMR algorithm that converges to an optimal solution within only a few iterations. The simulation results demonstrate the impact of system parameters and verify the correctness and effectiveness of the OEMR algorithm.

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